

Lecture 1: Linear Transformations & Matrices

Core Definitions

We begin by establishing the properties of fields mapping vector components.

Definition : Linear Map

A function $T : V \rightarrow W$ between two vector spaces over the same field F is called a **linear transformation** if it satisfies:

1. Additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$
2. Homogeneity: $T(cv) = cT(v)$ for all $c \in F$ and $v \in V$

Fundamental Theorems

Theorem : Rank-Nullity Theorem

Let V and W be vector spaces, where V is finite-dimensional. If $T : V \rightarrow W$ is a linear map, then:

$$\dim(\text{null } T) + \dim(\text{range } T) = \dim V$$

Homework Practice

Exercise : Identity Dimension Mapping

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find a structured matrix representation M_T and verify the output dimension explicitly:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$